

Reasoning LLMs

Module 1 — What they are, why now

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What do we mean by "reasoning"?

Recall — look something up:

| "What's the course code for Stanford's Transformers class?" → CME 295

Reasoning — work it out:

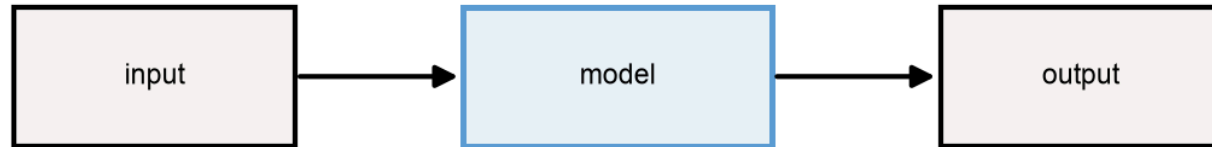
| "A bear was born in 2020. How old in 2025?" → $2025 - 2020 = 5$

Reasoning = solving a problem through a **multi-step process**, not a single lookup.

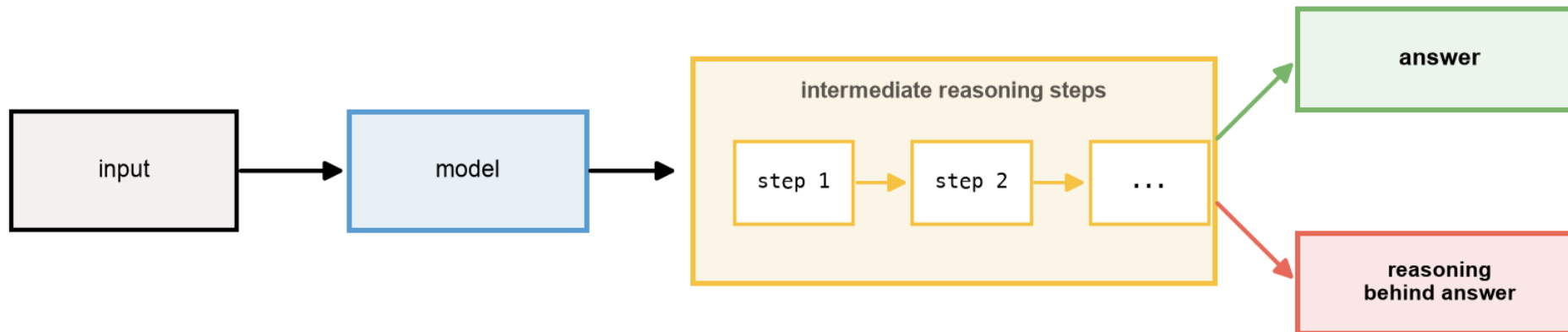
What is a reasoning LLM?

Reasoning LLM = reasoning chain + answer

Traditional LLM



Reasoning LLM



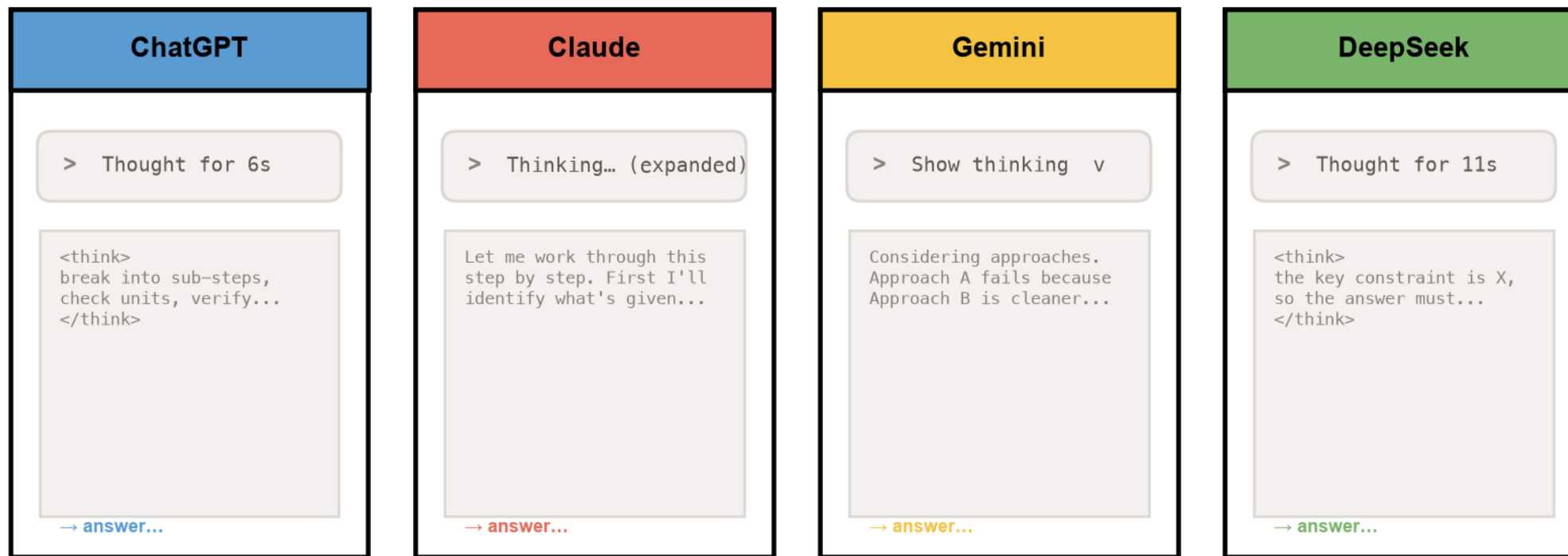
Output = reasoning chain + answer.

The output contract changes: **Output = reasoning chain + answer**

You've already seen these everywhere

Thinking shows up everywhere

illustrative mockups — not real screenshots



A catch: you pay for the thoughts

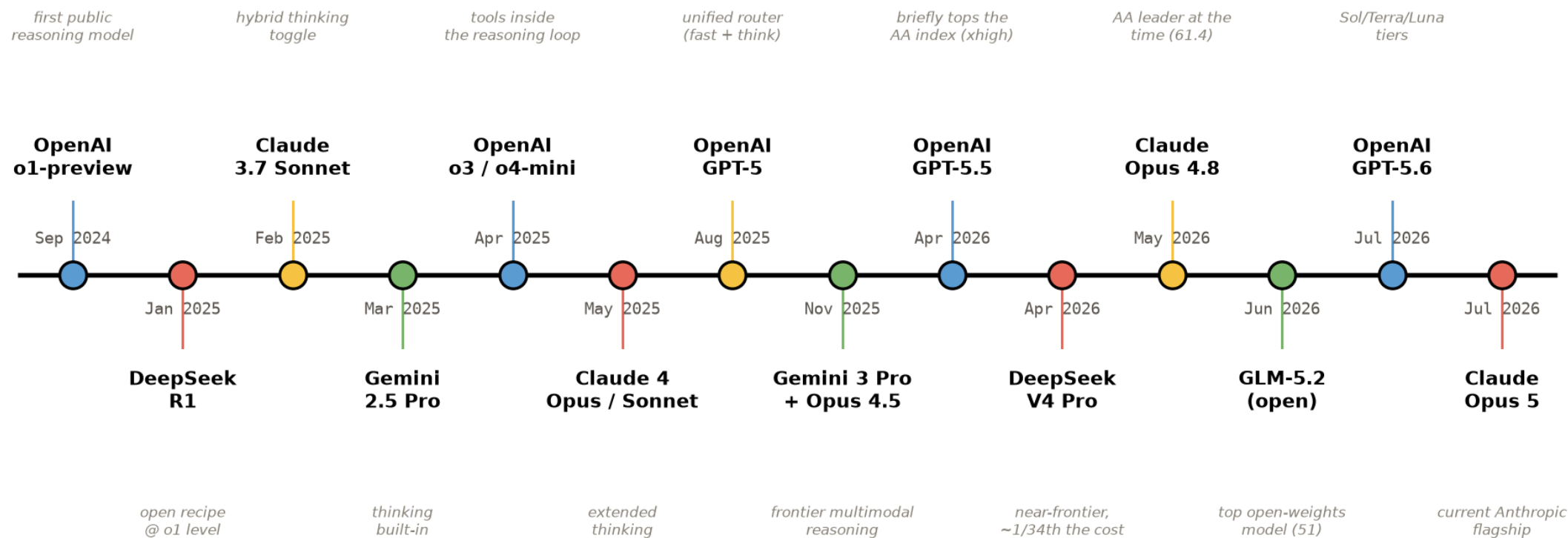
Reasoning tokens are **billed as output tokens** — on every major API.

- UIs show a *summarized* thought, not the raw chain (barely-readable, and exposed chains could train rivals)
- So there's a real incentive to get **the most reasoning per token**

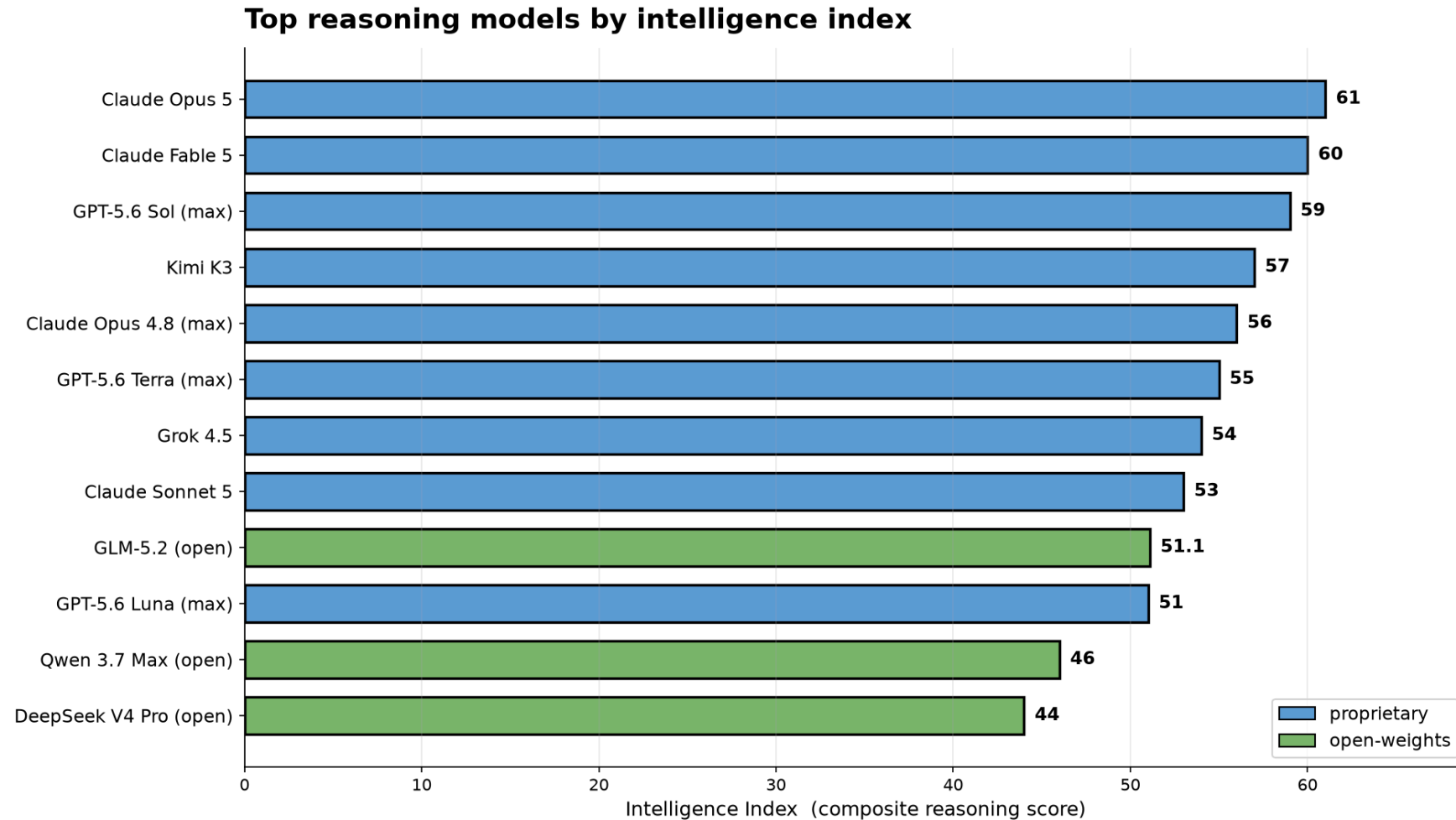
More on controlling the thinking budget later.

How we got here

The reasoning-model wave · Sep 2024 → Aug 2026



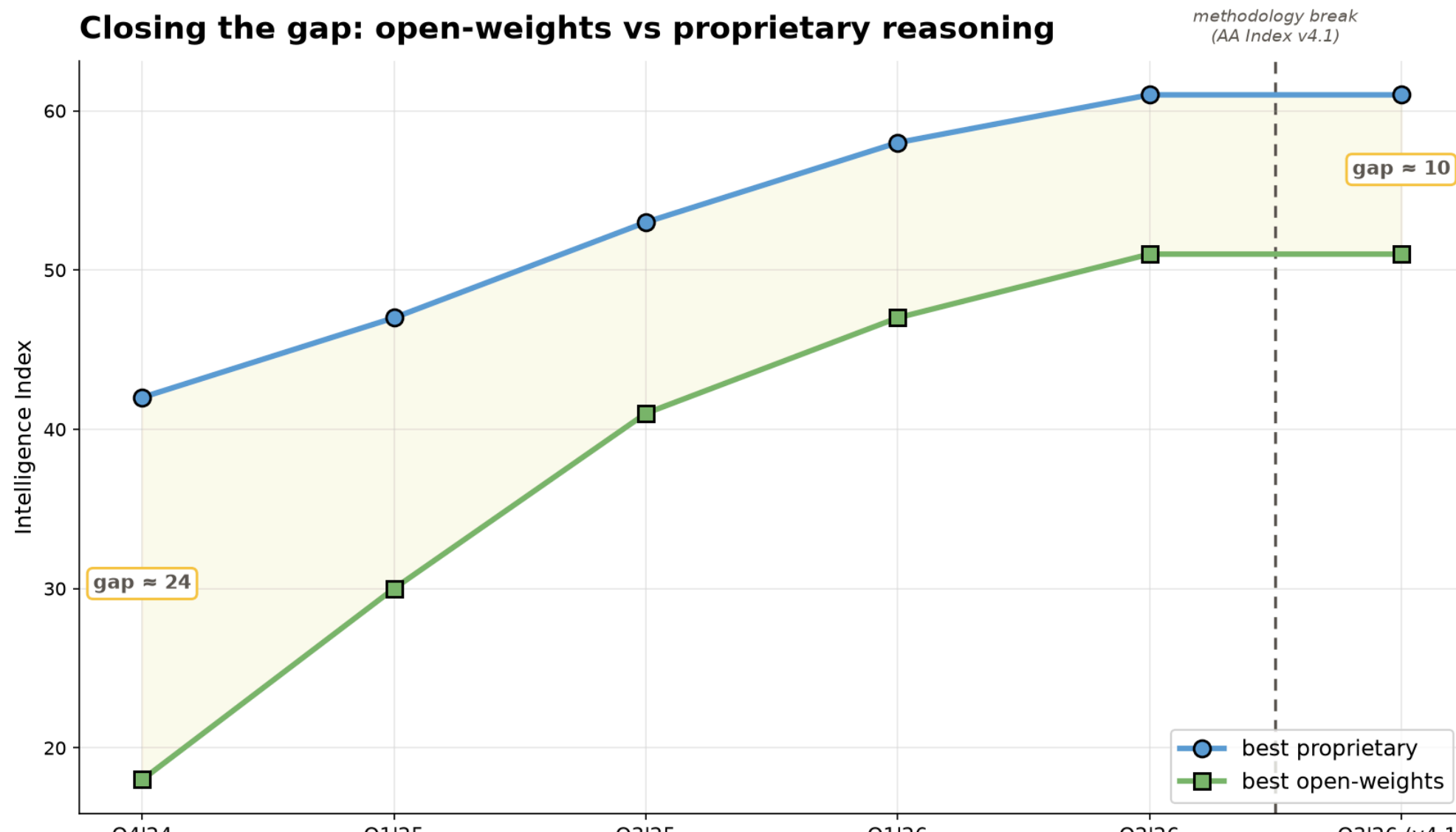
Where the models rank today



Source: Artificial Analysis Intelligence Index style · as of Aug 2026 (v4.1) · approximate, see notes

Demand concentrates on a handful of top labs (Artificial Analysis ~Aug 2026)

Closing the gap



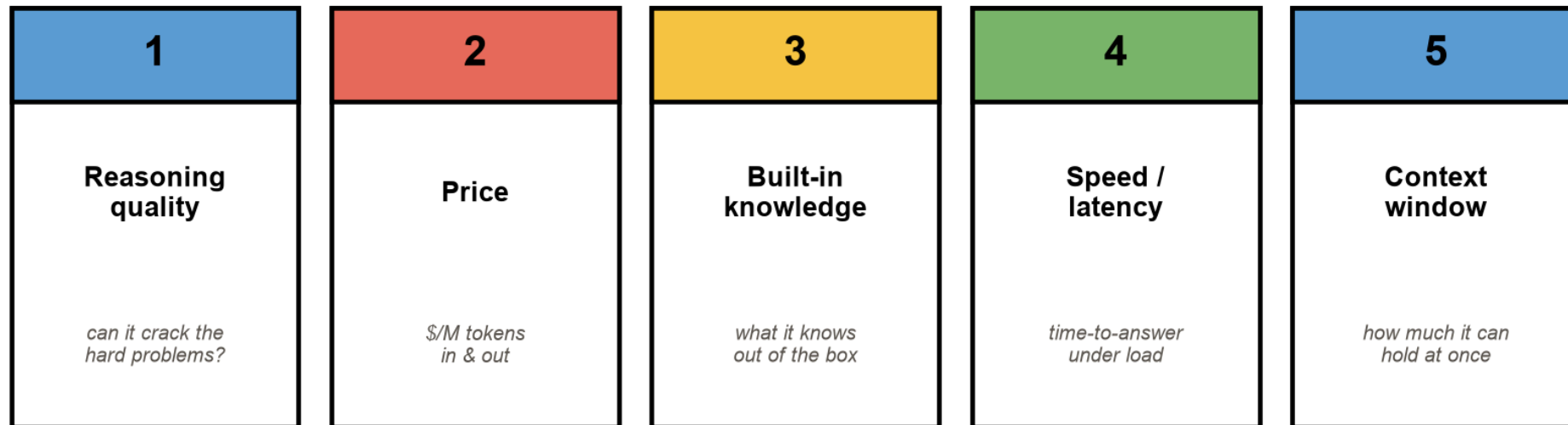
Why reach for a reasoning model?

It earns its extra tokens when the task is a **derivation**, not a lookup:

- Decomposes hard, multi-constraint problems
- Hallucinates less on problems it can *check itself* on
- Gives you the **reasoning**, not just the verdict

What actually matters when you pick one

Choosing a reasoning model — 5 criteria



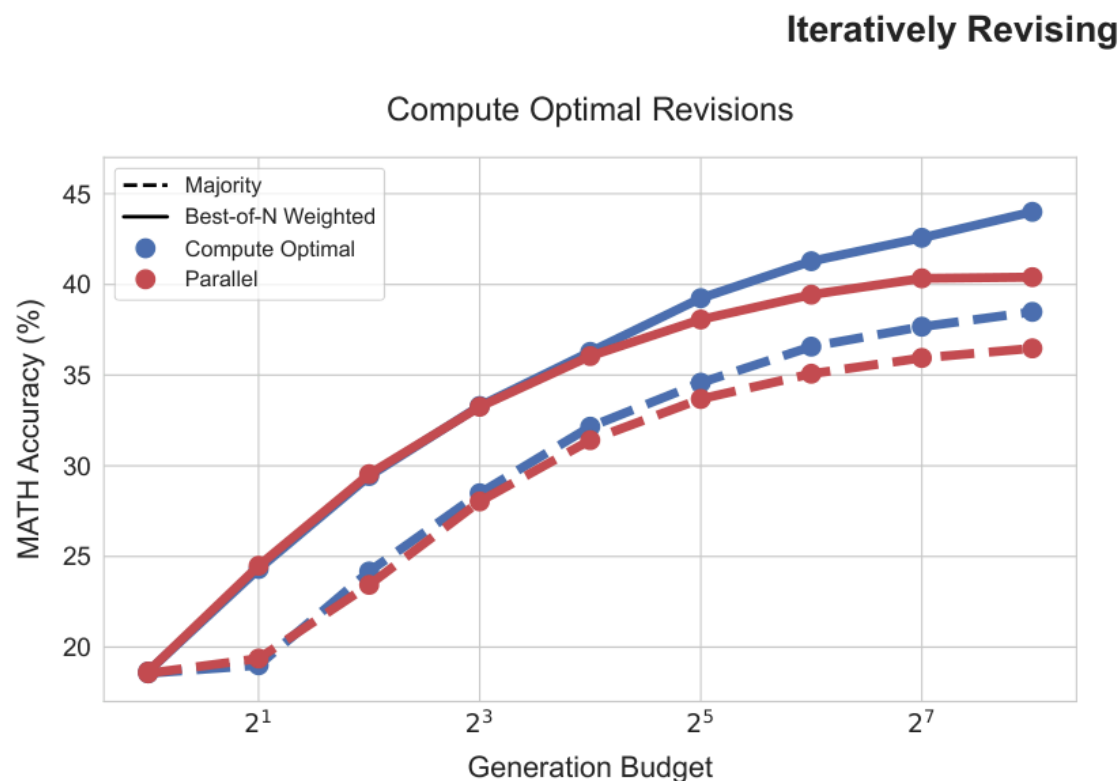
Reasoning quality · price · built-in knowledge · speed · context window.

Price is the other axis

Model (~Aug 2026)	\$/M out	
GPT-5.6	~\$30	frontier
Claude Opus 5	~\$25	frontier
Gemini 3.1 Pro	~\$12	
DeepSeek V4 Pro	~\$0.87	near-frontier, ~34× cheaper

The expensive model isn't always the right call. (*Approx; we test this in notebook 08.*)

Why does thinking longer help?



Every generated token is another forward pass — reasoning tokens **buy compute** before the model commits.

The simplest version of the trick

Chain-of-thought prompting elicits reasoning

recreation — after Wei et al., 2022 (arXiv:2201.11903)

Standard prompting	Chain-of-thought prompting
Q: Roger has 5 tennis balls. He buys 2 more cans of 3 balls each. How many does he have now? A: The answer is 11. (no working shown) X	Q: Roger has 5 tennis balls. He buys 2 more cans of 3 balls each. How many does he have now? A: Roger started with 5 balls. 2 cans of 3 balls each is 6 balls. $5 + 6 = 11$. The answer is 11. /
no reasoning -> wrong	reasoning steps -> right

"Let's think step by step." Chain-of-thought, scaled up, is the core idea. *(after Wei et al., 2022)*

Two reasons a reasoning chain helps

1. **Decomposition** — break a problem the model has never seen into sub-problems it *has*.
2. **More compute** — more tokens before the answer = more computation spent on it.

Extended chains and self-correction ("wait, that's wrong...") **emerge from RL**, not just from prompting.

When to use one — and when not to

When to reach for a reasoning model

GOOD FIT	POOR FIT
+ Complex, single-shot problems + Fewer hallucinations needed + Diagnosis-style reasoning + Explanations & step-by-step + Hard problems other LLMs fail	- Speed / latency-critical tasks - Cheap bulk extraction - High-volume summarization - Simple lookups & formatting - Tight per-call cost budgets

How to prompt a reasoning model

- Keep the prompt **short and direct** — don't hand-write the chain of thought for it
- Frame it as an **expert generalist**; state the goal, not the steps
- Give it **clear success/failure criteria** for what a good answer looks like
- Tune **effort/budget** to the task instead of always maxing it

You're not prompting a model — you're prompting an agent

Prompt it like an agent, not a model

Goal	What you ultimately want accomplished.
Return format	Exact structure / schema of the output.
Warnings	Edge cases, what to avoid, constraints.
Context	Background, data, prior state.
Tools	web search · code interpreter · retrieval
Actions	Steps it may take to reach the goal.

You're prompting an agent now.

The effort knobs

- **OpenAI:** `reasoning.effort = none / low / medium / high / xhigh / max`
- **Anthropic:** `output_config.effort = low / medium / high / xhigh / max`
(current models) — `thinking.budget_tokens` still on Haiku 4.5
- **Google:** thinking config on the Gemini models

Same model, dial the depth to the task.

Limitations (keep them honest)

- 5–50× more tokens → **cost & latency**
- Early/open models: **language mixing**, sensitivity to few-shot
- The visible "thought" is a summary — and **may not be faithful**
(Anthropic: "reasoning models don't always say what they think")

Live demo →

`notebooks/01_foundations_reasoning_and_cot.ipynb`

A tiny model learns that **a scratchpad beats answering directly** — the whole idea, measured, on your laptop.

Recap

- Reasoning $\neq$ retrieval — **output = reasoning + answer**
- You can see it (the "Thinking..." tag) and you *pay* for it
- Trade tokens for accuracy — only when the task needs it
- Next: how DeepSeek **trained** one, then we **rebuild** it in code